Troubleshooting Chart 7-2.
Starting Circuit Problems: Engine Fails to Crank

Note: Before jumping out solenoid terminals, completely vent the engine compartment, especially with gasoline engines, since sparks will be created.

Turn on some lights powered by the cranking battery and try to crank the engine. Do the lights *go out?* **NO**	**YES** If the solenoid makes a rapid "clicking," the battery is probably dead; replace it. If the solenoid makes one loud click, the starter motor is probably jammed or shorted; free it up or replace it as necessary. Before taking the starter out, try turning the engine over by placing a wrench on the crankshaft pulley nut. If the engine won't turn this is the problem; there may be water in the cylinders (page 382) or the engine may be seized up.
When cranking, do the lights *dim?* **NO**	**YES** Check for voltage drop from poor connections or undersized cables. Try cranking for a few seconds, then feel all connections and cables in the circuit—battery and solenoid terminals, and battery ground attachment on the engine block. If any are warm to the touch, they need cleaning. If this fails to show a problem, use a voltmeter as shown in Figure 7-25, then try to crank the engine. If there is no evidence of voltage drop, check for a jammed or shorted starter.
Is the ignition switch faulty? **NO** **TEST:** With a jumper wire or screwdriver blade, bridge the battery and ignition switch terminals on the solenoid. If the starter motor now cranks, the ignition circuit is defective.	**YES** Replace ignition switch or its wiring as needed.
Is the solenoid defective? **NO** **TEST:** Use a screwdriver blade to jump out the two heavy-duty terminals on the solenoid. (This procedure is tricky: Observe all precautions outlined in the accompanying text.) If the starter now spins, the solenoid is defective.	**YES** Replace the solenoid.
Is there full battery voltage at the starter motor when cranking? **NO** **TEST:** Check with a multimeter between the starter positive terminal and the engine block when cranking.	**YES** The starter is open-circuited and needs replacing. First check its brushes for excessive wear or sticking in their brush holders.
No voltage: The battery isolation switch is probably turned off!	

you will never forget it. It is not a bad idea to sniff them before starting any other detective work—this may rapidly narrow the field of investigation.

Preliminary tests. A couple of quick steps will isolate problems in the starting circuit. The solenoid will have two heavy-duty terminals: one is attached to the battery's positive cable; the other is attached with a second cable (or short strap in the case of preengaged starters) to the starter itself. There will also be one or two small terminals. If only one of the small terminals has a wire attached to it, this is the one we need. If both small terminals have wires attached, one is a ground wire; we need the *other one* (it goes to the ignition switch). If in doubt, turn off the ignition and battery switches and test from both solenoid terminals to ground with an ohmmeter (R × 1 scale on an analog meter). The ignition switch wire should give a small reading; the ground wire will read 0 ohms.

Turn the battery isolation switch back on. Now bypass the ignition switch circuit (and neutral start switch if fitted) by connecting a jumper wire or screwdriver blade from the *battery terminal* on the solenoid to the *ignition switch* terminal (Figures 7-24A, 7-24B, and 7-24C). If the motor cranks, the ignition switch or its circuit is faulty. If the solenoid clicks but nothing else happens, the battery may really be dead (check it again), or the starter is probably bad. (But note that the solenoid on a preengaged starter will sometimes throw the pinion into engagement with the flywheel but then fail to energize the motor due to faulty main points, so *always check*

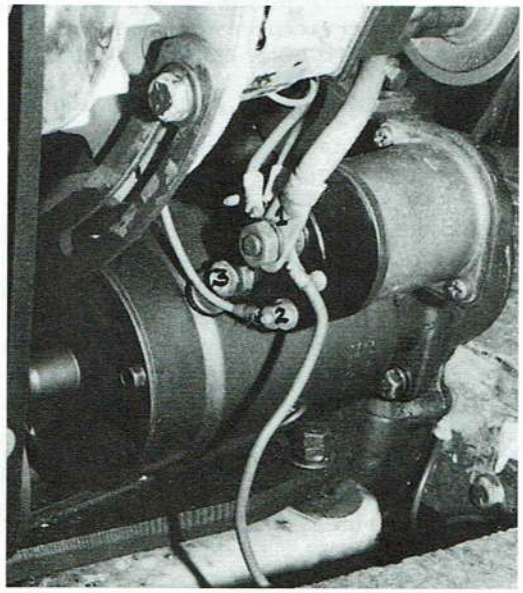


FIGURE 7-24A. A typical preengaged starter motor and solenoid. Jumping across 1 and 2 bypasses the ignition switch; 1 and 3 bypasses both the ignition switch and the solenoid. The starter should spin but will not engage the engine's flywheel.